A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

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1 Introduction

Irradiation-assisted stress corrosion cracking (IASCC) is a major degradation mode of austenitic stainless steel core internals in light-water reactors [1]. In irradiated stainless steels exposed to high-temperature water, oxidation of the grain boundary weakens it and precedes intergranular crack initiation [2], and grain-boundary oxidation has been identified as a necessary precursor of IASCC at low doses [3]. In 304 stainless steel (304SS) exposed to pressurized water reactor (PWR) primary water, the extent of this oxidation depends on the chemical changes introduced by prior irradiation. Deng et al. [4] found that increasing the proton irradiation dose produced deeper intergranular oxidation in 304 nuclear-grade stainless steel after water exposure, accompanied by a decrease in the Cr content of the intergranular oxide. Cr depletion is not the only chemical change that matters. Using model alloys with Si and Cr contents chosen to mimic the segregated boundary, Du et al. [5] showed that Si enrichment promotes intergranular oxidation and cracking even when Cr is not depleted: Si forms SiO_2 that is unstable in high-temperature water and dissolves preferentially, leaving a porous oxide that provides pathways for oxygen diffusion along grain boundaries. These observations connect irradiation to both the chemistry and the penetration depth of the intergranular oxide. Turning them into a predictive relationship requires a description of how the changing oxide composition and structure affect oxygen transport and, consequently, the advance of the oxidation front.

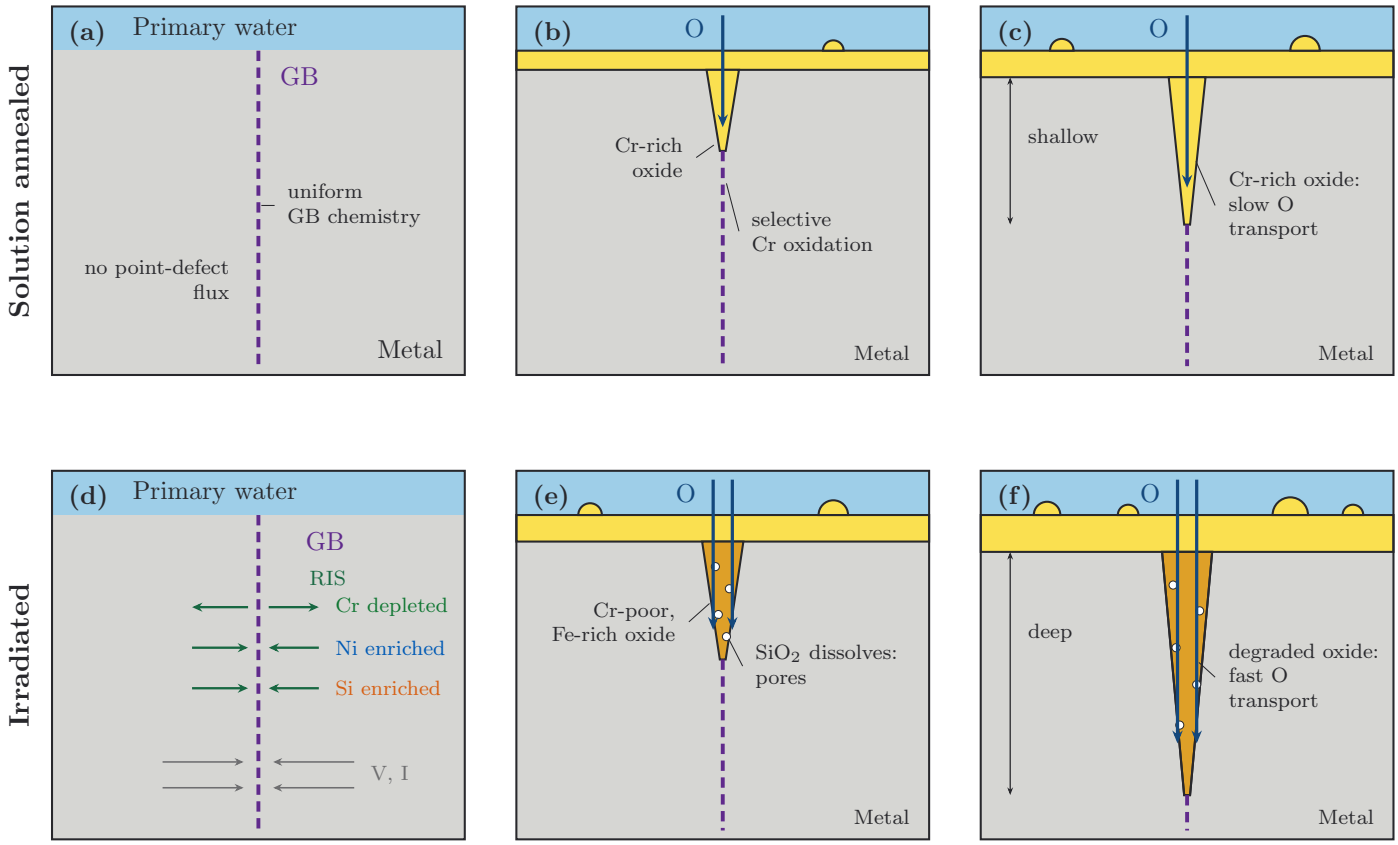


Figure 1: Physical picture of intergranular oxidation in primary water without (a–c) and with (d–f) prior irradiation. (a) Solution-annealed boundary with uniform chemistry. (b) Oxygen entering at the boundary mouth oxidizes Cr selectively and forms a Cr-rich intergranular oxide. (c) The Cr-rich oxide limits oxygen transport along the boundary and the penetration stays shallow. (d) Irradiation drives point defects to the boundary and the resulting radiation-induced segregation depletes Cr and enriches Ni and Si. (e) On exposure the Cr-depleted boundary forms a Cr-poor, Fe-rich oxide, and the segregated Si oxidizes and dissolves, leaving pores. (f) The degraded oxide transports oxygen faster along the boundary and the oxide penetrates deeper.

Radiation-induced segregation (RIS) establishes the local alloy composition from which the intergranular oxide develops. Coupling between solute and irradiation-generated point-defect fluxes redistributes alloying elements near grain boundaries. Measurements on irradiated 304 nuclear-grade stainless steel reveal Cr depletion and Ni enrichment at the grain boundary [4], and atom-probe tomography further resolves Si enrichment alongside Ni [6]. Subsequent oxidation selectively consumes alloying elements, changing both the oxide composition and the adjacent metal, where Ni enrichment has also been observed [6]. The lower Cr content of the oxide formed on irradiated boundaries has been associated with reduced resistance to oxygen diffusion [4]. Oxide protectiveness therefore has a direct transport meaning: an oxide that restricts oxygen diffusion more strongly limits the oxygen available for further oxidation.

Si oxidation and dissolution alter this transport resistance by changing the structure through which oxygen moves. The role of Si identified by Du and co-workers was established with model alloys of systematically varied Cr and Si contents in hydrogenated water, which allowed the contributions of Cr depletion and Si enrichment to be separated. They attributed the greater oxide degradation and cracking susceptibility at higher Si content to dissolution of SiO_2 in the inner oxide and the resulting porous, defective film, and their STEM analysis of intergranular oxides indicated faster grain-boundary diffusion at higher Si content [5]. Related microscopy on proton-irradiated 316L identified Si enrichment at oxide tips and porosity behind them, consistent with preferential Si oxidation followed by dissolution [**kuang2022**]. Although 316L differs from 304SS in bulk composition, this observation is used here as mechanistic support rather than as a quantitative reference; the same oxidation and dissolution of segregated Si appears as a step in the IASCC mechanism summarized by Was et al. [2]. Together, these studies provide a basis for including dissolution-generated free volume in the transport description. With the dependence on Cr content, they motivate representing the evolving oxide composition and free volume through a single effective oxygen diffusivity, which connects the segregation of multiple elements to a common quantity governing oxygen transport.

Each half of this sequence has been modeled separately. Models coupling solute and point-defect transport predict the composition redistribution produced by irradiation. Kadambi and co-workers developed a phase-field model for segregation at grain boundaries and dislocation structures in austenitic Fe–Cr–Ni [7], and Oplinger and co-workers formulated a rate-theory frame-

work for the Cr enrichment–depletion transition at grain boundaries in Fe–Cr [8]. These models take the alloy and irradiation conditions as input and return the segregation profile, but they stop at the chemical state of the boundary. On the oxidation side, the dependence of intergranular penetration depth on dose and exposure time has been characterized experimentally, but the boundary chemistry enters existing kinetic descriptions only through fitted coefficients. To our knowledge, no model connects the segregated composition to the advance of the oxidation front through a transport quantity computed from both. Establishing this connection requires tracking how the segregated alloy supplies the interfacial reactions, how those reactions change the oxide, and how the resulting diffusivity affects subsequent oxygen delivery. Segregation profiles, oxide composition, interfacial Ni enrichment, and penetration depth then provide complementary observables for assessing the coupling.

Here, we develop a coupled RIS and grain-boundary oxidation model for 304SS, represented by the Fe–Cr–Ni–Si system. The model links defect-mediated solute redistribution and selective interfacial consumption to an effective oxygen diffusivity, D_{eff}, determined by the evolving oxide composition and free volume. Within this description, increasing the Cr content of the oxide lowers the effective diffusivity, whereas Si oxidation followed by complete dissolution contributes free volume that enhances transport. The same D_{eff} enters both oxygen transport along the grain boundary and oxygen delivery across the oxide to the metal interface. The model therefore connects RIS to oxidation kinetics through the resulting change in oxygen transport, with metal consumption feeding back on the local composition. The complete-dissolution treatment is a simplification motivated by the mechanism discussed by Du and co-workers [5]. We examine the dose and exposure-time dependence of oxide penetration under sequential irradiation and water exposure, using separate calibration of the RIS and oxidation components and a common set of oxidation parameters across doses. The framework also permits predictions of concurrent RIS and oxidation within the same transport description.

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